

Graph Neural Network Tutorial – 2

1. Message passing graph neural networks (MPGNN) can be mathematically represented in different ways as given below. Which of the following statements are correct?

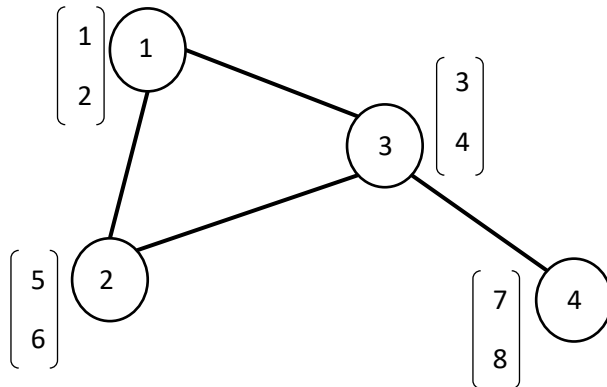
[Standard notation is used]

- A. $H^{(k)} = \sigma \left(AH^{(k-1)}W_{neigh}^{(k)} + H^{(k-1)}W_{self}^{(k)} \right)$
- B. $h_u^{(k)} = \sigma \left(W_{self}^{(k)}h_u^{(k-1)} + \left(\sum_{v \in N(u)} W_{neigh}^{(k)}h_v^{(k-1)} \right) + b^{(k)} \right)$
- C. $H^{(k)} = \sigma \left((A + I)H^{(k-1)}W_{shared}^{(k)} \right)$

$$H^{(k)} \in \mathbb{R}^{v \times d^{(k)}}, \quad H^{(k-1)} \in \mathbb{R}^{v \times d^{(k-1)}}, \quad b^{(k)} = b_{self}^{(k)} + b_{neigh}^{(k)}, \quad W_{shared}^{(k)} \in \mathbb{R}^{d^{(k)} \times d^{(k-1)}}$$

$$W_{self}, W_{neigh} \in \mathbb{R}^{d^{(k)} \times d^{(k-1)}}$$

- I. (A) equation is a graph level (i.e., matrix format) representation of MPGNN and (B) equation is a node level representation of MPGNN.
 - II. When $d^{(k)} = d^{(k-1)}$ and $W_{self} = W_{neigh}^{-1}$, (A) equation is equivalent to (C) equation.
 - III. In (B), $W_{neigh}^{(k)}h_v^{(k-1)}$ represents the message from neighborhood node v.
 - IV. Relu is the only activation function that can be used in MPGNN.
 - V. Using multiple dense layers (without activations) on the self-node instead of just one dense layer (without activations) always result in better performance.
2. With the given graph with feature vectors, a single layer MPGNN is used for a node classification task. Activation function is not used in this MPGNN. Calculate the final node embedding from the MPGNN layer for the node number 3.



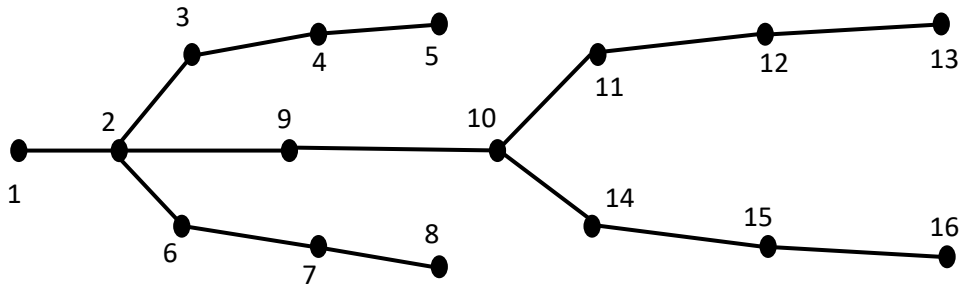
$$W_{neigh} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$$

$$W_{self} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- I. [48, 27]

- II. [45, 23]
 III. [24, 45]
 IV. [0, 0]
 V. [42, 42]
3. Graph convolution network (GCN) is a popular GNN architecture used in many GNN application domains. There are multiple variants of GCN architectures. Which of the following equations represents such GCN operations?
 [Standard notation is used]
- I. $h_v^{(l)} = \sigma(W_l x^{(l)} + b^l)$
 II. $H^{(k)} = \sigma(D^{-1/2} A D^{-1/2} H^{(k-1)} W^{(k)})$
 III. $H^{(k+1)} = \sigma(D^{-1} A H^{(k)} W_k^T + H^{(k)} B_k^T)$
 IV. $h_v^{(l)} = \sigma(\sum_{u \in N(v)} \alpha_{uv} W^{(l)} h_u^{(l-1)})$
 V. $h_v^{(l)} = \sigma(W^{(l)} \cdot \text{CONCAT}(h_v^{(l-1)}, \text{AGG}(\{h_u^{(l-1)}, \forall u \in N(v)\}))$
4. When training a GNN, over-smoothing effect adversely affect the performance of the model. Which of the following statements are true,
- I. Over-smoothing is when all node embeddings converge to the same value
 II. When number of GNN layers are increased in a GNN architecture, receptive field of a node is also increased. This in turn increase the overlap of receptive fields of two nodes resulting in over-smoothing effect.
 III. Over-smoothing can be successfully mitigated by using regularization methods like dropout, batch-normalization and label smoothing.
 IV. Over-smoothing can be successfully mitigated by adding skip connections between GNN layers.
 V. Over-smoothing effect generally happens after adding more than 512 GNN layers.
5. In Graph neural networks, under-reaching is an effect that adversely affect the performance of a model. This happens when information from distant nodes doesn't reach the given node(s). Since one GNN layer aggregates information from 1-hop neighbors, to aggregate information from K-hop neighbors, K number of GNN layers are needed. Thus, what are the possible values we can use for K (i.e., number of layers) to mitigate the under-reaching in the given graph?



- I. 6
 II. 4
 III. 8
 IV. 10
 V. 2